

Symbolic methods of mathematical analysis

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Chapter 1.

Ring of power series

1.1. Power series

A natural generalization of the notion of a polynomial is the notion of a power law. number of rows. Let t be a symbolic variable and k be a field, then an expression of the form

$$\sum_{n=0}^{\infty} a_n t^n = a_0 + a_1 t + a_2 t^2 + \dots, a_n \in k \quad (1.1)$$

it is called a power series or, more generally, a series in powers of t . At the same time $a_0, a_1, \dots$ they are called the coefficients of the series. For a series, the following values are considered to be different: if they differ by at least one coefficient.

Under the sum of two rows

$$\sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} b_n t^n$$

second row

$$\sum_{n=0}^{\infty} a_n + b_n t^n,$$

and under the product of rows

$$\sum_{n=0}^{\infty} a_n t^n \cdot \sum_{n=0}^{\infty} b_n t^n$$

second row

$$\sum_{n=0}^{\infty} c_n t^n,$$

the coefficients of which are calculated using the formula

$$c_n = a_n b_0 + a_{n-1} b_1 + \cdots + a_0 b_n$$

which is exactly the same as the one used for calculating coefficients of the product of two polynomials. Both when adding and when multiplication to calculate the coefficient for a given degree t is sufficient. perform a finite number of actions accurately.

Remark. If desired, you can understand the power series as infinite a sequence of numbers from k and instead of a series (1.1) write $a_0, a_1, a_2, \dots$. Two such sequences are added element-wise, and multiplied by a tricky rule:

$$a_0, a_1, a_2, \dots \cdot b_0, b_1, b_2, \dots = c_0, c_1, c_2, \dots$$

exactly

$$c_n = \sum_{k=0}^n a_{n-k} b_k = \sum_{k=0}^n a_k b_{n-k}$$

Example 1. Find the square of a row

$$1+t+t^2+t^3+\dots$$

Note that all its coefficients are equal to 1 and therefore in the formulas for c_n it should be put in

$$c_n = \sum_{k=0}^n 1 = n+1$$

Therefore

$$(1+t+t^2+t^3+\dots)^2 = 1+2t+3t^2+\dots$$

or

$$\sum_{n=0}^{\infty} t^{n^2} = \sum_{n=0}^{\infty} (n+1)t^n$$

The set of all series in powers of t with coefficients from the field k with so the entered arithmetic operations are denoted as $k[[t]]$ This is about the value is also used in Sage, only Sage already knows that it is a set ring.

sage: var('t')	1...
t...	2...
sage : QQ [[t]]	3...
Power Series Ring in t over Rational Field	4...

If you truncate all members of the series whose degree is greater than or equal to n , then the series will turn out to be a polynomial of degree n . We will call this function the operation is truncated by a row and denoted using parentheses. For example,

$$[3+t^2+t^5+\dots]_3 = 3+t^2$$

In Sage, this is written as

sage: (3+t^2+t^5).truncate(3)	5...
3+t^2	6...

The definitions of the sum and product of power series are arranged as follows, for calculating the coefficient for t^n in total, it is enough to know the coefficients by t^n for summands, so

$$[f+g]_n = [f]_n + [g]_n$$

to calculate the coefficient for t^n there are enough rows in the product know the coefficients for $1, t, \dots, t^n$ for cofactors, so

$$[f \cdot g]_n = [f \cdot [g]_n]_n = [[f]_n \cdot g]_n = [[f]_n \cdot [g]_n]_n$$

These formulas allow us to reduce the proof of series theorems to remain on polynomials.

Theorem 1. The set of all series in powers of t with coefficients from the fields k with arithmetic operations so entered is a ring.

The proof of this theorem is reduced to checking the axioms from the definition of use of the ring. Let us check for example the 4th axiom associativity of multiplication-

$$(f \cdot g) \cdot h = f \cdot (g \cdot h)$$

To do this, it is sufficient to prove that the coefficient for t^{n-1} matches on the left with the coefficient at t^{n-1} on the right. But the truncation of the left part is equal to

$$[(f \cdot g) \cdot h]_n = [[f \cdot g]_n \cdot h]_n = [[f_n \cdot g_n]_n \cdot h]_n = [(f_n \cdot g_n) \cdot h]_n$$

and the truncation of the right part is equal to

$$[f \cdot (g \cdot h)]_n = [f_n \cdot g_n \cdot h_n]_n = [f_n \cdot (g_n \cdot h_n)]_n$$

These truncations are equal, since multiplication of polynomials is associative. But then the coefficients for $1, t, \dots, t^{n-1}$ to the left and right of the proved ra- the circumstances match. Due to the randomness of the choice of n , the above is true for all coefficients.

Traditionally staidIt is written starting from the lowest member of the group. on, that is, a term with a nonzero coefficient having the smallest degree of stump. The degree of this term in Sage is called evaluation. For well- in the left row, this value is assumed to be +00:

```

sage: Qd [[ t ]] (3 * t^6 + t^2 + t^5) (3 * t^6 + t^2 + t^5).valuation()
2...
sage: QQ [[ t ]] (0).valuation()
+ Infinity

```

7...
8...
9...
10...

The set of all power series whose score is greater than or equal to n , is the ideals of the ring $k[[t]]$ and is denoted as $O(t^n)$ For the proof- It is necessary to check the fulfillment of two axioms from the definition of an ideal.

Two power series, say f and g , are equal modulo $O(t^n)$ if

$$f-g \in O(t^n).$$

Traditionally, instead of

$$f=g\text{mod } O(t^n)$$

they write

$$f=g+O(t^n).$$

Looking at our definition of truncation, it's not hard to see that

$$f = [f]_n + O(t^n)$$

Since there is no way to work directly with the infinite- In Sage, instead of the power series f , we consider a class is created $f_n + O(t^n)$ that is, the element of the quotient ring $k[[t]] / O(t^n)$

When we multiply two rows

$$f = f_n + O(t^n) \quad g = g_n + O(t^n)$$

as elements of the quotient ring, we get

$$f_n g_n + O(t^n) = f_n g_n + O(t^n)$$

As we have already noted, the coefficients of the product fg by $1, t, \dots, t^{n-1}$ match the coefficients for the same powers $f_n g_n$. Now

we can- would it be possible to prove this by referring to the consistency of actions in the source code? ring and factor ring. However, we have already used this in our previous documentation. because the set of power series is a ring.

The same statement is true for addition. Therefore, re- progress from power series to elements of a quotient ring $k[[t]] / O(t^n)$ allows perform arithmetic operations for a finite number of actions and gives the result that has the correct coefficients for powers of t which are less than n . Therefore, we are said to work with series with precision up to terms of order n , and the value of n itself is called precision.

If f is a polynomial of $k[t]$, then $k[[t]](f)$ is considered in Sage as a power series with a finite number of nonzero terms. Record $k[[t]](f).0(n)$ turns it into an element of the quotient ring $k[[t]] / O(t^n)$. So, to find the square of a row from Example 1 in Sage, you need to set this row with with some accuracy, that is, as an element of a quotient ring modulo $O(t^n)$ and then square it:

```
sage: QQ[[t]](sum(t~n for n in range(10))).0(5)~2
```

11.

```
1+2*t+3*t-2+4*t-3+5*t-4+0(t-5)
```

12.

1.2. A series that is the inverse of the specified one

Let the set be a series of

$$f = f_0 + f_1 t + f_2 t^2 + \dots$$

Then the power series is

$$x = x_0 + x_1 t + x_2 t^2 + \dots$$

satisfying the equation

$$fx = 1$$

call it the inverse of f and write $x = f^{-1}$

Theorem 2. If the power series is not equal to zero when $t = 0$ that is available the inverse of such a power series is.

Proof. We will search for a series that is the inverse of this one, not in the original one in the ring of power series, and in the quotient rings $k[[t]] / O(t^n)$

$$(f + O(t^n)) \cdot (x + O(t^n)) = 1 + O(t^n)$$

By $n = 1$ the statement is obvious:

$$f = f_0 + O(t) \quad \overline{x} = \frac{1}{f_0} + O(t) \quad f \cdot \overline{x} = 1 + O(t)$$

Let's assume that we have found a solution for some n , that is, such as $x_0, \dots, x_{n-1}$ what

$$(f + O(t^n))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + O(t^n)) = 1 + O(t^n)$$

We will look for a solution in quotient rings $k[[t]] / O(t^{n+1})$ by the method of undetermined the calculated coefficient, assuming

$$\overline{x} = x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + u t^n + O(t^{n+1})$$

where u is an undefined value. Coefficients of the product fx by $1, \dots, t^{n-1}$ they coincide with the coefficients at the same degrees for the pro- issues

$$(f + O(t^n))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + O(t^n)) = 1 + O(t^n)]$$

that is, the coefficient at 1 is one, and the rest are zero. Coef- ficient for t^n in the work

$$(f+O(t^{n+1}))(x_0+x_1t+\cdots+x_{n-1}t^{n-1}+ut^n+O(t^{n+1}))$$

equal to the sum of the coefficient for the same degree in

$$(f+O(t^{n+1}))(x_0+x_1t+\cdots+x_{n-1}t^{n-1}+O(t^{n+1}))$$

and the f0u value. Equating this coefficient to zero, we unambiguously we define and and thereby find the solution of the equation $fx = 1$ in factor- the ring $k[[t]] / O(t^{n+1})$

Starting with $n = 1$ we will consistently find the values $x_1, x_2, \dots$. and as a result, we get the following series

$$x = \frac{1}{f_0}+x_1t+x_2t^2+\dots$$

what is the coefficient for t^m if fx is zero, that is

$$fx = 1$$

This proves the theorem.



The proof of the theorem was almost constructive: we can find clearly the coefficient for any given degree t, but we can't continue the process goes on indefinitely. Therefore, we can present x with accuracy up to members t^n where n is some large number. For example,

$$\text{s\~{a}ge:}Q[[t]](1-t)\text{bad hbox}-1$$

13.

$$\begin{array}{l} 1+\underline{1}+\underline{u}\geq 2+\underline{u}\geq 3+\underline{u}\cdot \underline{-4}+\underline{u}\cdot \underline{5}+\underline{u}\cdot \underline{7}+\underline{1}\cdot 7+\underline{5}\cdot 8+\underline{t}\quad 14.\\ -9+t^{-10}+t^{-11}+t^{-12}+t^{-13}+t^{-14}+t^{-15}+t^{-16}+t^{-17}+t^{-18}+t^{-19}+0(t-20)\\ \text{sage: } QQ[[t]](1-t+t-3)\sim -1\quad 15\dots\\ 1+z+t^{-1}2-z^{-4}4^{-2}=z^{-1}+5-z^{-2}6^{-5}-5^{-7}+1z^{-8}+3\quad 16\dots\\ \sim 9^{-4}*t-10+3*t-11-4*t-13-7*t-14-t=15^{-}\\ 3*t?16+4*t-17+11*t-18+14*t-19+0(t-20) \end{array}$$

In Sage by default $n = 20$ In the documentation, this value is called accuracy (precision):

```
sage: QQ[[t]].default_prec() 17.
20. 18.
```

If desired, you can redefine it by defining the ring in more detail: `sage: PowerSeriesRing(QQ, t, default_prec=5)` 19.

Power Series Ring in t over Rational Field 20.

```
sage: PowerSeriesRing(QQ, t, default_prec=5)(1-t)^-1
```

```
21 (1-t)^-1 1 + t + t^2 + t^3 + t^4 + O(t^5) 22...
```

In some cases, this infinite process can be described in a finite way. then write the answer in the original power-law ring. rows. For example,

$$f = 1-t$$

then

$$1-tx_0+x_1t+x_2t^2+\dots=1$$

means that

$$x_0+x_1-tx_0+x_2-tx_1t^2+\dots=1$$

This gives

$$x_0=1, x_1=x_0, x_2=x_1, \dots$$

Therefore

$$(1-t)^{-1} = 1+t+t^2+t^3+\dots$$

As a corollary of the proved theorem, the ideal $O(t)$ is formed by all powers of rows that do not have inverse elements in the ring $k[[t]]$ Remark. Ring $k[[t]]$ is an integral ring, so you can determine the field of quotients of this ring, in which any element is invertible. non-zero element.

1.3. Differentiation of a power series ring

In Alegbra's course, the differentiation D of the ring $k[t]$ was introduced:

$$D: \sum_{n=0}^N f_n t^n \rightarrow \sum_{n=1}^N n f_n t^{n-1}$$

Let's continue its continuation on $k[[t]]$ by accepting

$$D: \sum_{n=0}^{\infty} f_n t^n \rightarrow \sum_{n=1}^{\infty} n f_n t^{n-1}$$

This rule defines a linear mapping of the power series ring to itself. To call it a differentiation of the ring of power series, we need also prove that it satisfies the Leibniz rule

$$D(fg) = gDf + fDg$$

To this end, we compare the coefficients for t^n on the left side of this equality with the coefficient for the same degree on the right side. Differentia- this reduction reduces the degrees by one, so this coefficient will not change the value of if we truncate f, g to terms t^{n+2} . But for polynomials f_{n+2} and $[g]_{n+2}$ The Leibniz formula has already been proved in the course of algebra, and therefore these coefficients are the efficients match.

For a row

$$f = f_0 + f_1 t + f_2 t^2 + \dots$$

the value is defined only at one point when $t = 0$

$$f(0) = f_0$$

At any other point, the question of the meaning of the series leads us to the question of convergence of the series.

Theorem 3 (of Taylor). Let

$$f = f_0 + f_1 t + f_2 t^2 + \dots$$

then

$$f_n = \frac{1}{n!} D^n f(0)$$

10...

Proof. If we differentiate this series, we get

$$Df = f_1 + 2f_2 t + 3f_3 t^2 + \dots$$

From here we can find the derivative at zero: $Df(0) = f_1$. Differentiating once again, we get

$$D^2 f = 2f_2 + 6f_3 t + \dots$$

where from $D^2 f(0) = 2f_2$ etc.

□

1.4. The equation $Dx = ax$ and the exhibitor

The equation connecting the desired seta term or series with its derivative it is called a differential equation. Most of the models in the fur- both physics and mathematics are described by such differential equations. The simplest of the differential equations is

$$Dx = ax \quad a \in k$$

Remark. This equation with a negative real a describes various processes of cooling and decay, when the rate of decrease of x is proportional to x . For example, when describing the cooling of a liquid poured into a container, you can take x as the temperature of the liquid taken relative to ambient temperature, and assume that the rate of temperature change is proportional to temperature, i.e. Dx is proportional to x . Then hotter liquid, the faster it cools down.

The equation $Dx = ax$ is not solved in polynomials, since the degree of Dx is one less than the power of x . However, it is easily solved in power-law equations. Substituting the expression

$$x = x_0 + x_1 t + x_2 t^2 + \dots$$

in the differential equation, we obtain

$$\dot{x}_1 + 2x_2 t + 3x_3 t^2 + \dots = ax_0 + ax_1 t + ax_2 t^2 + \dots$$

This results in recurrent relations to the coefficients:

$$x_1 = ax_0, x_2 = \frac{a^2}{2}x_0, x_3 = \frac{a^3}{3!}x_0, \dots$$

It is not difficult to get explicit formulas from here:

$$x_n = \frac{a^n}{n!}x_0 = \frac{a^n}{n(n-1)\dots 1}x_0 = \frac{a^n}{n!}x_0$$

Thus,

$$x = x_0 \left(1 + at + \frac{(at)^2}{2!} + \dots \right)$$

is the solution of the differential equation $Dx = ax$. In this way, so, we have a solution in rows that contains one indeterminate a constant x_0 which we will interpret as the value of x when $t = 0$.

When we divide by n , we assume, what's in our field $n \neq 0$ what doesn't hold in Galois fields. From this location, we work in $\mathbb{Q}$ or its extensions of $\mathbb{R}$, $\mathbb{C}$.

This simple reflection leads to several important conclusions: lyrics.

Definition 1. Power series

$$1 + t + \frac{t^2}{2!} + \dots = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

called the exponent and denoted as $\exp(t)$.

The expression $\exp(at)$ for any $a \in k$ we can consider as a step foam row. In Sage, for example, $\exp(3t)$ can be set as:

```
sage: QQ[[t]] (3*t). exp ()
```

23...

$$1 + 3t + \frac{9}{2}t^2 + \frac{27}{2}t^3 + \frac{27}{8}t^4 + \frac{81}{40}t^5 + \dots$$

$$+ \frac{81}{80}t^6 + \frac{243}{560}t^7 + \frac{729}{4480}t^8 + \frac{43}{4480}t^{10} + \frac{2187}{492800}t^{11} + \frac{2187}{1971200}t^{12} + \frac{6561}{2560000}t^{13} + \dots$$

$$19683 / 358758400 * 1.44 + 19683 / 1793792000 * 1^{-15} + \\ 049 / 28700672000 * 1.6 + 177147 / 487911424000 * t^{-2} + 59049 / 97582848000 * t^{-18} + 177147$$

Theorem 4. Differential equation

$$Dx = ax$$

has a solution equal to x_0 by $t = 0$

$$x = x_0 \exp (at)$$

Other solutions in the ring $k[[t]]$ equals x_0 by $t = 0$ this is the differential- this equation does not have.

Proof. Expression

$$x = x_0 \sqrt{1+at+\frac{(at)^2}{2!}+ \dots}$$

is a solution because

$$Dx = x_0 D\sqrt{1+at+\frac{(at)^2}{2!}+ \dots} = x_0 a + \frac{at^2}{2!} + \dots = ax$$

If it had a different solution

$$x = x_0 + x_1 t + x_2 t^2 + \dots$$

then, substituting it in $Dx = ax$ we would get

$$x_n = \frac{a}{n} x_{n-1} = \frac{a^2}{n(n-1)} x_{n-2} = \dots = \frac{a^n}{n(n-1) \dots 1} x_0 = \frac{a^n}{n!} x_0$$

which means that this solution coincides with the one already specified. □

Two very important corollaries can be drawn from the proved theorem.

Theorem 5 (addition theorem). For any $a, b \in k$ right

$$\exp (at) \exp (bt) = \exp ((a+b)t)$$

Proof. Let's put

$$x = \exp (at) \exp (bt),$$

then by virtue of the Leibniz rule

$$Dx = D \exp (at) \cdot \exp (bt)+ \exp (at) \cdot D \exp (bt)$$

By virtue of theorem 4

$$D \exp (at) = a \exp (at) \quad D \exp (bt) = b \exp (bt)$$

so

$$Dx = (a+b) \exp (at) \cdot \exp (bt) = (a+b)x$$

Because $x(0) = 1$ by virtue of the uniqueness of the solution of the equation $Dx = ax$ established in theorem 4,

$$x = \exp ((a+b)t)$$

what and tr. d. □

In Sage, for example, $\exp(3t)$ can be set as:

```
sage: 2[[t]](t).exp()*Q[[t]](2*t).e
1 +3*t+9/2*t-2+9/2*t-3+27/8*t-4 81/40*t~5 + 26
81/80*1%+76+243/560*1^7+729/4480+1^8
243/4480*-9+729/44800*t-10+2187/492800*t +
2187/1971200*t-12+6561/25625600*1-13+
19683/358758400*b^-14+19683/1793792000*1.15 +
049/28700672000*1^16+177147/487911424000*E^
+ 59049/975822848000*t~18 + 177147/18540634112000*
t~19+0(t~20)
```

```
sage: QQ[[t]] (3*t). exp ( ) 27...
```

$$1+3*\pi+9/2*t \geq 2+9/2*5^{-2}+27/8*t^{-2}+81/40*z^{-2} \quad s = \quad 28$$
$$81/80*1^{\circ}6+243/560*1^{\circ}7+729/4480*t^{-8}+$$
$$243/4480*t-9+729/44800*t-10+2187/49280 \quad +$$

$$2187 / 1971200 * t^{-12} + 6561 / 25625600 * t^{-13} - 19683 / 358758400 * t^{-14} + 19683 / 1793792000 * t^{-15} + \\ 049 / 28700672000 * t^{-16} + 177147 / 487911424000 * t^{-2} + 59049 / 975822848000 * t^{Red18} + 177147 / 185406341120$$

Theorem 6. For any $a \in k$ right

$$\exp (at)^{-1} = \exp (-at)$$

Proof. The statement follows from the addition theorem when $b = -a$

$$\exp (at) \exp (-at) = \exp ((a-a)t) = \exp (0) = 1$$

□

1.5. The equation $D^2x = a^2x$ and hyperbolic functions

Let us now turn to the simplest equation relating the second problem. the derivative, that is, acceleration, and x, namely

$$D^2x = a^2x$$

Substituting in this equation the expression

$$x = \sum_{n=0}^{\infty} x_n t^n$$

we get

$$\sum_{n=0}^{\infty} n(n-1)x_n t^{n-2} = a^2 \sum_{n=0}^{\infty} x_n t^n$$

This results in recurrent relations to the coefficients:

$$n(n-1)x_n = a^2x_{n-2}$$

It is not difficult to get explicit formulas from here:

$$x_n = \frac{a^2}{n(n-1)}x_{n-2} = \frac{a^4}{n(n-1)(n-2)(n-3)}x_{n-4} = \dots$$

If n is even, then for $n / 2$ steps we will come to x_0

$$x_n = \frac{a^n}{n!} x_0 .$$

If n is odd, then we arrive at x_1

$$x_n = \frac{a^{n-1}}{n!} x_1$$

Thus,

$$x = x_0 \left(1 + \frac{(at)^2}{2!} + \frac{(at)^4}{4!} + \dots \right) + \frac{x_1}{a} \left(at + \frac{(at)^3}{3!} + \frac{(at)^5}{5!} + \dots \right)$$

is the solution of the differential equation $Dx = ax$ In this way, so, we have a solution in rows containing two constants x_0 and x_1 which are easily interpreted as x and Dx values when $t = 0$

Definition 2. Power series

$$t + \frac{t^3}{3!} + \frac{t^5}{5!} + \dots$$

called the hyperbolic sine and denoted as $\sinh (t)$ a power law row

$$1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \dots$$

called the hyperbolic cosine and denoted as $\cosh (t)$.

In Sage, for example, $\sinh (t)$ can be set as:

```
sage: QQ[[t]] (t). sinh()
```

29...

$t + 1/6*t^3 < 3+1 /$

$$+1 / 39916800 * t^{-11} + 1 / 622702000 * t^{-13} + 1 / 13076743680000 * t^{15} + 1 / 3566874280900 * t^{-17} +$$

$$1 / 121645100408832000 * t^{-19} + 0(t^{-20})$$

Theorem 7. Differential equation

$$D^2 x = a^2 x$$

equipped with initial conditions

$$x(0) = x_0 \quad Dx(0) = x_1$$

has a solution

$$x = x_0 \cosh(at) + \frac{x_1}{a} \sinh(at)$$

Other solutions satisfying the same initial conditions in the ring $k[[t]]$ this differential equation has no.

Looking at the rows for $\sinh t$ and $\cosh t$, it's hard not to notice the chunks of the row for exponents. Row

$$\cosh t = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \dots$$

consists of even terms of the series for the exponent, so

$$\cosh t = \frac{\exp(t) + \exp(-t)}{2}$$

Row

$$\sinh t = t + \frac{t^3}{3!} + \frac{t^5}{5!} + \dots$$

consists of odd terms of the series for the exponent, so

$$\sinh t = \frac{\exp(t) - \exp(-t)}{2}$$

These formulas are called Euler formulas. They show that for the solution of a second-order differential equation is sufficiently ex-ponents.

1.6. The equation $D^2x + a^2x = 0$ and trigonometric functions

Let us now turn to the equation

$$D^2x + a^2x = 0$$

This is a very popular mathematical model called the linear OS. with a cillator.

Substituting in this equation the expression

$$x = \sum_{n=0}^{\infty} x_n t^n$$

we get

$$\sum_{n=0}^{\infty} n(n-1)x_n t^{n-2} = -a^2 \sum_{n=0}^{\infty} x_n t^n$$

This results in recurrent relations to the coefficients:

$$n(n-1)x_n = -a^2 x_{n-2}$$

It is not difficult to get explicit formulas from here:

$$x_n = -\frac{a^2}{n(n-1)}x_{n-2} = +\frac{a^4}{n(n-1)(n-2)(n-3)}x_{n-4} = \dots$$

If n is even, then for $n / 2$ steps we will come to x_0

$$x_n = \frac{(-1)^{\frac{n}{2}} a^n}{n!} x_0$$

If n is odd, then we arrive at x_1

$$x_n = \frac{(-1)^{\frac{n-1}{2}} a^{n-1}}{n!} x_1$$

Thus,

$$\bar{x} = x_0 \left(1 - \frac{(at)^2}{2!} + \frac{(at)^4}{4!} - \dots \right) + \frac{x_1}{a} at \left(1 - \frac{(at)^2}{3!} + \frac{(at)^4}{5!} - \dots \right)$$

is the solution of the differential equation $Dx = ax$ In this way, so, we have a solution in rows containing two constants x_0 and x_1 which are easily interpreted as x and Dx values when $t = 0$

Definition 3. Power series

$$t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots$$

called the sine and denoted as $\sin(t)$ a power series

$$1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots$$

called the cosine and denoted as $\cos(t)$.

In Sage, for example., $\sin(t)$ you can set it like this:

sage: QQ[[t]] (t). sin() 31...

$t^{-1}/6 * t^{-3} + 1/120 * t^{-5} - 1/5040 * t^{-7} + 1/362880 * t^{-9} - 1/39916800 * t^{-11} + 1/622702080 * t^{-13} =$ 32...

$1/1307674368000 * t^{-15} + 1/35568742809600 * t^{-17} -$

$1/121645100408832000 * t^{-19} + O(t \sim 20)$

Theorem 8. Differential equation

$$D^2x + a^2x = 0$$

equipped with initial conditions

$$x(0) = x_0 \quad Dx(0) = x_1$$

has a solution

$$x = x_0 \cos(at) + \frac{x_1}{a} \sin(at)$$

Other solutions satisfying the same initial conditions in the ring $k[[t]]$ this differential equation has no.

If you work on $\mathbb{C}$, you can take $a = ib$ and reduce the differential equation $k D^2x = b^2x$ previously reviewed. At the same time

$$\cos t = \cosh it = \frac{\exp(it) + \exp(-it)}{2}$$

And _

$$\sin t = \frac{1}{i} \sinh ut = \frac{\exp(it) - \exp(-it)}{2i}$$

These formulas are called Euler formulas. They show that for solutions of this second-order differential equation also suffice. but the exhibitors, but you will have to pay for this by going to the complex the plane.

1.7. Other differential equations

Differential equation

$$F(D^n x, D^{n-1} x, \dots, x, t) = 0$$

it can be solved in power series. However, in practice, the same problem occurs a problem that we have already encountered when dealing with a power series. We can describe a procedure that, from the solution found with exactly- t^n will give the solution up to terms t^{n+1} . We can apply this procedure many times and find a solution with some accuracy- but we can't turn the formulas for recalculating the new coefficient into The coefficient is passed through the old ones to an explicit formula for calculating the coefficient. This trick only works for a very narrow range of differential equations. equations.

Sage has a built - in function for solving the equation

$$Df = a(t)f + b(t)$$

where a and b are polynomials or series with powers of t.

Example 2. Find a solution to the initial problem

$$Df = 2-3t+4t^2 f+3-4t^2 \quad f(0) = 3$$

Let's define the coefficients a, b of the equation and apply method `solve_linear_de`:

```
sage: a = QQ[t] (2-3 * t+4 * t^2)                                     33 33
sage: b = QQ[[t]](3-4 * t^2)                                         34...
sage: a.solve_linear_de(b=b, f0=3 prec=15) c = 15 3 + 9*t 1+8/2 *t-2-10/73 *t-3+95/24 *t-4+4
~5 - 1301/720 *t^-6 -3359/2520 *t^-7 +158701/40320
~8 + 18721/36288*t~9 - 5845291/3628800*t~10 + 258607/259200*t~11 + 32265269/43545600*t~12 -
173190679/283046400*t~13 + 61616509/1585059840*t
Δ14+0 (t ~ 15)                                                     3636
```


Homework

Theoretical tasks

- 1) How are arithmetic operations defined in the ring of power series $k[[t]]$
- 2) Prove that the set of power series is a ring?
- 3) Prove that the set $O(t^2)$ is an ideal of the power-law ring $k[[t]]$
- 4) What is the evaluation of a power series?
- 5) What is the precision of a power series?
- 6) Under what conditions does the series have an inverse?
- 7) Define the expt series, $\sinh t$ $\sin t$.
- 8) Formulate an addition theorem for exponentials.
- 9) Using Euler's formula, prove:
 - a) the basic hyperbolic identity $\cosh^2 t - \sinh^2 t = 1$
 - b) basic trigonometric identity $\sin^2 t + \cos^2 t = 1$

Practical tasks

- 1) Let

$$f = \sum_{n=0}^{\infty} t^n, g = \sum_{n=0}^{\infty} nt^n$$

- a) Find the coefficient for t^5 in the expression $f+g$
- b) Find the coefficient for t^5 in the expression fg
- c) Find the coefficient for t^5 in the expression fg
- d) Find the first 15 terms of the series for $1/f$

2) Calculate the power series estimate

$$\overline{1-\frac{t}{2}} \cos t$$

3) Truncate the row $\sin^3 2t$ before t^5

4) Find the first 10 terms of the series $\tan t = \sin t / \cos t$

5) Above field A, find the solution to the initial problem

a) $Dx+2x = 0 \quad x(0) = 0$

b) $Dx+2x = 0 \quad x(0) = 3$

c) $D^2x+2x = 0 \quad x(0) = 0 \quad Dx(0) = 0$

d) $D^2x+2x = 0 \quad x(0) = 1 \quad 1, \quad Dx(0) = 3$

e) $D^2x-2x = 0 \quad x(0) = 1 \quad Dx(0) = 3 \quad 3,$

6) Over the field C, express the solutions of the previous problem in terms of the exponential you.

Chapter 2.

The exhibitor

2.1. Numerical experiments with the exponent

In the last chapter, we found a solution to the differential equation

$$Dx = x$$

as a formal power series

$$\exp t = 1+t+\frac{t^2}{2!}+\cdots = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

(2.1)

If we substitute any value for t, say, $t = 1$ then semi-
the sum of a series of rational numbers is given, namely

$$\exp 1 = 1+1+\frac{1}{2!}+\cdots = \sum_{n=0}^{\infty} \frac{1}{n!}$$

This expression is called the sum of a
number series or simply numeric nearby.

We don't know how to split an infinite number of numbers, but
we can substitute in place of infinity, various large integers:

```
sage: sum([1/factorial(n) for  
n in range(10)]) 98641/36288
```

37...
3838

```
sage: sum([1/factorial(n) for n in  
range(20)]) 82666416490601/30411275102208
```

39...
40...

To compare the resulting fractions, convert them to decimals:

```
sage: RR(sum([1/factorial(n) for n in
```

```
range(10)])
```

```
2.71828152557319
```

```
sage: RR(sum([1/factorial(n) for
```

```
range(20)])
```

```
2.71828182845905
```

```
sage: RR(sum([1/factorial(n) for
```

```
range(40)])
```

```
2.71828182845905
```

```
sage: RR(sum([1/factorial(n) for
```

```
range(100)])
```

```
2.71828182845905
```

It is easy to see that with 20 terms or more, the same number is obtained. Of course, if you increase the number of bits allocated to the decimal fraction, you may notice a slight difference in these amounts:

```
sage: RealField(70)(sum([1/factorial(n) for n in
range(20)]))
```

```
2.7182818284590452349
```

```
sage: RealField(70)(sum([1/factorial(n) for n in
range(30)]))
```

```
2.7182818284590452354
```

```
sage: RealField(70)(sum([1/factorial(n) for n in
range(42)]))
```

```
2.7182818284590452354
```

So we can't calculate an infinite sum exactly, but with a small error, we can write

$$e \approx 2.718281828459045235360287471352 \dots$$

For other values of t , we will see approximately the same thing. For example, for $t = 2$

```
sage: RealField(70)(sum([2^n/factorial(n) for n in
range(20)]))
```

7.3890560989301740963 56...

sage: RealField(70)(sum([2-n/factorial(n) for n in
range (30)]) 57...

7.3890560989306502272 58...

sage: RealField(70)(sum([2~n/factorial(n) for n in
range (42)])) 59...

7.3890560989306502272 60...

This results in approximately the same number:

$$\exp 2 \approx 7.38905609893065022723042313404.$$

This allows us to consider the power series of the exponent as a function in in the sense of school mathematics, namely, as a rule, which puts in the context of matching a number t is a number

$$\sum_{n=0}^{\infty} \frac{t^n}{n!}$$

2.2. The Cauchy Criterion

The basic idea behind real number theory is that next: if something can be calculated with any given precision, then this something exists. An example of such an approach is the following: we now need the Cauchy criterion.

Let there be a series of numbers

$$a_0, a_1, a_2, \dots$$

from $\mathbb{R}$ or $\mathbb{C}$ and let for any positive number $\epsilon > 0$ there is such a thing N , which amounts to

$$\sum_{n=0}^M a_n$$

by $M \geq N$ they differ from each other by no more than ϵ . In that case

there is a number a such that the sums

$$\sum_{n=0}^M a_n$$

by $M \geq N$ they differ from this number by no more than ϵ .

In other words, if the sums differ from each other by arbitrarily if it is not enough, then you can specify a number that differs from these production sums. not much at ease.

For convenience, we will break down the Cauchy criterion into parts. If for any positive number $\epsilon > 0$ there is an N such that the sums of

$$\sum_{n=0}^M a_n$$

by $M \geq N$ If they differ from each other by no more than ϵ , then we say that row

$$a_0 + a_1 + a_2 + \dots$$

it fits. If for any positive number $\epsilon > 0$ there is such an N , what are the amounts

$$\sum_{n=0}^M a_n$$

by $M \geq N$ if they differ from the number A by no more than ϵ , then we say that the series

$$a_0 + a_1 + a_2 + \dots$$

converges to the number A and write

$$a_0 + a_1 + a_2 + \dots = A \quad \dots = A$$

The Cauchy criterion states that a convergent series necessarily converges to some number.

As a consequence of these definitions, we indicate a very simple necessary convergence criterion.

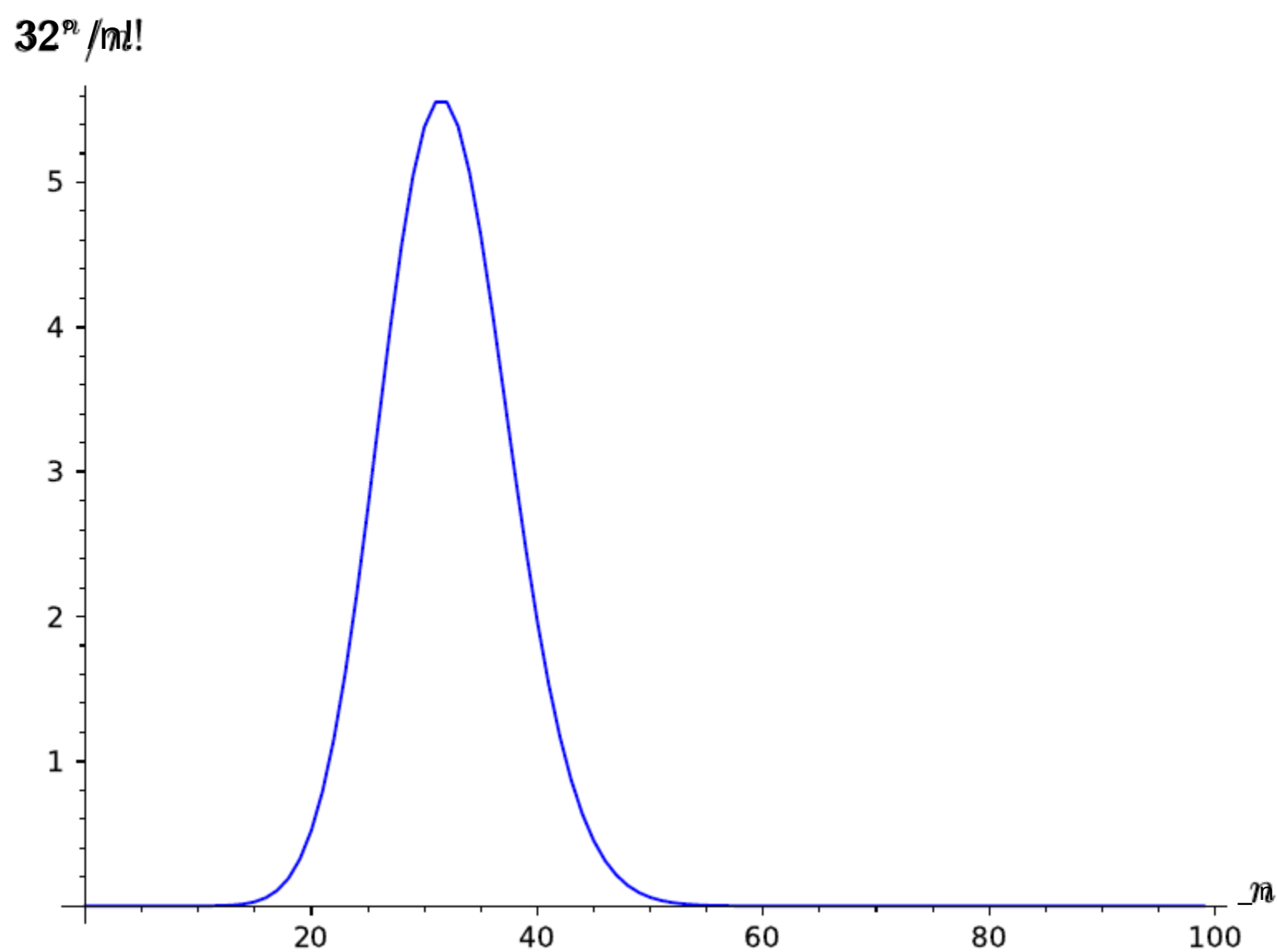


Figure 2.1. Graph of the convertibility of the terms of the exponential series at $t = 32$ from member number, scale by its ordinate — 10^{12}

Theorem 9 (necessary convergence sign, Euler sign). If the row is

$$a_0 + a_1 + \dots$$

it fits, that a_n as n grows, it becomes arbitrarily small.

Proof. By condition for any $\epsilon > 0$ there is an N such that

$$\sum_{n=0}^M a_n, \sum_{n=0}^{M+1} a_n, (M > N)$$

they don't depend on each other for more than ϵ . But the difference between these amounts is a_{M+1} . Therefore, for any ϵ , there is an N such that $|a_M| < \epsilon$ only just $M > N$ □

Often, in a practical account, the sum of several terms is calculated a row and noticing that the next member is too small or, on the contrary, too large. If the range is too large, they rush to draw a conclusion about the convergence or divergence of the series. These judgments may be wrong.

Example 3. Members of the exponent series when $t = 32$ when changing $n = 0$ before 30 grow rapidly and very radically, reaching values of the order of 10^{12} However, this does not prevent

2.3. Geometric progression series

Number series

$$1, q, q^2, q^3, \dots$$

it is called a geometric progression. When n increases infinitely, q^n tends to infinity or zero, depending on whether it is greater than $|q|$ units or less. By the Euler criterion, the series

$$1 + q + q^2 + \dots$$

diverges when $|q| \geq 1$

Let's turn to the case of $|q| < 1$ Because

$$(1 + q + \dots + q^{M-1})(1 - q) = 1 - q^M$$

right

$$1 + q + \dots + q^{M-1} = \frac{1 - q^M}{1 - q}$$

Hence the inequality

$$|1 + q + q^2 + q^3 + \dots + q^{M-1}| < \frac{1}{1 - q} \tag{2.2}$$

It is essential for us that the right-hand side of this inequality does not depend on M .

Let N be a natural number and the numbers M_1, M_2 greater than N , and $M_2 > M_1$ then the difference is

$$1 + q + \dots + q^{M_1-1}$$

and _

$$1 + q + \dots + q^{M_2-1}$$

can be evaluated modulo as

$$q^{M_1} + \dots + q^{M_2-1} = q^{M_1} (1+q+\dots) < \frac{q^{M_2}}{1-q} \leq \frac{q^N}{1-q}$$

So this estimate is independent of M_1 and M_2

Since when $|q| < 1$ value q^n tends to zero, for any $\epsilon > 0$ you can choose an N such that

$$\frac{q^N}{1-q} \leq \epsilon$$

Hence, by the Cauchy criterion, there exists such a real number A and, what

$$1+q+\dots+q^{M-1}$$

differs from A arbitrarily small for all sufficiently large M , then there is

$$1+q+q^2+q^3+\dots = A$$

Of course, from

$$1+q+\dots+q^{M-1} = \frac{1-q^M}{1-q}$$

you can immediately see that

$$1+q+q^2+q^3+\dots = \frac{1}{1-q}$$

In other words, the series of geometric progressions converges when $|q| < 1$ and its amount is equal to

$$1+q+q^2+q^3+\dots = \frac{1}{1-q}$$

Example 4. Let's illustrate the convergence of calculations when $q = \frac{1}{3}$

sage: $q = 1 / 3$	61...
sage: $RR(1/(1-q))$	62...
1.500000000000000	63...
sage: $RR(\text{sum}([q^n \text{ for } n \text{ in range}(10)]))$	64...
1.49997459736829	65...
sage: $RR(\text{sum}([q^n \text{ for } n \text{ in range}(20)]))$	666

1.49999999956980	67...
sage: RR(sum([q-n for n in range(30)]))	68 68
1.499999999999	69...
sage: RR(sum([q^n for n in range(42)]))	70...
1.50000000000000	71...

Remark. In the last chapter, we saw that in the ring $\mathbb{Q}[[t]]$ reverse to to the polynomial $1-t$ there will be a row $1+t+t^2 + \dots$ Now we can see that this series converges to $\frac{1}{1-t}$

2.4. The e-number

As a basic oneFor an example, consider a number series

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots \quad (2.3)$$

which is obtained from the power series $\exp t$ when $t = 1$ Let's mark it as your own- It's a good idea.

1. The series (2.3) converges. To prove this, we will use the estimate

$$\frac{1}{n!} < \frac{1}{1 \cdot 2 \cdot 2 \dots 2} = \frac{1}{2^{n-1}}$$

Let N be a natural number and the numbers M_1, M_2 greater than N , and $M_2 > M_1$ then the difference is

$$\sum_{n=0}^{M_1} \frac{1}{n!}$$

And _

$$\sum_{n=0}^{M_2} \frac{1}{n!}$$

so

$$\sum_{n=M_1+1}^{M_2} \frac{1}{n!}$$

can be evaluated modulo as

$$\sum_{n=M_1+1}^{M_2} \frac{1}{2^{n-1}} = \frac{1}{2^{M_1}} (1 + \frac{1}{2} + \dots) < \frac{1}{2^N} (1 + \frac{1}{2} + \dots)$$

Applying the estimate (2.2) when $q = \frac{1}{2}$ we have

$$\frac{1}{2^N} \frac{1}{1 - \frac{1}{2}} = \frac{1}{2^{N-1}}$$

Obviously, for any $\epsilon > 0$ there is an N such that

$$\frac{1}{2^{N-1}} \leq \epsilon$$

With this choice of N and any M_1 and M_2 difference

$$\sum_{n=0}^{M_1} \frac{1}{n!}$$

And

$$\sum_{n=0}^{M_2} \frac{1}{n!}$$

there will be less ϵ . This means that the sum (2.3) converges to a certain value, a real number. This number is called the number e in honor of Euler and interpreted as the value of the power series $\exp t$ when $t = 1$

2. The number e lies between 2 and 3:

$$2 < e < 3$$

The first inequality immediately follows from the definition:

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots > 2$$

to prove the first inequality, it is sufficient to evaluate the terms of the series top as

$$\frac{1}{n!} < \frac{1}{1 \cdot 2 \cdot 2 \dots 2} = \frac{1}{2^{n-1}}$$

then

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots < 1 + 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

31...

the row on the left is a geometric one with $q = 1 / 2$ so

$$e < 1 + \frac{1}{1-\frac{1}{2}} = 3$$

3. The number e can be more accurately estimated as follows:

$$\sum_{n=0}^N \frac{1}{n!} < e < \sum_{n=0}^N \frac{1}{n!} + \frac{2}{(N+1)!}$$

For proof, we note that the remainder of the series

$$\sum_{n=N+1}^{\infty} \frac{1}{n!} = \frac{1}{(N+1)!} \left(1 + \frac{1}{N+2} + \frac{1}{(N+2)(N+3)} + \dots \right)$$

you can estimate from above as

$$\frac{1}{(N+1)!} \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots \right) = \frac{1}{(N+1)!} \frac{1}{1-\frac{1}{2}} = \frac{2}{(N+1)!}$$

For example, to calculate e up to 10^{-2} it is enough to take $N = 5$ members:

```
sage: 2/factorial (4+1).n() 4+1 72...
0.016666666666666 73...
sage: 2/factorial (5+1).n() 5+1 74...
0.0027777777777777 75...
```

In this case, the number e can be estimated as follows:

```
sage: sum([(1/factorial(m)).n() for m in range(5+1)
1 ) 76...
2.716666666666667 77...
sage: sum([(1/factorial(m)).n() for m in range(5+1)
] )+2/factorial 5+1 78...
2.719444444444444 79...
```

In particular the weaker statement follows:

$$e = 2.71 \pm 0.01$$

You can also calculate more correct digits in the same way. The e constant is usually protection in CAS:

sage: e.n() 80...

2.71828182845905 81...

sage: sum([(1/factorial(m)).n() for
m in range(17)]) 82 2.71828182845904 83...

sage: sum([(1/factorial(m)).n() for
m in range(18)]) 84 2.71828182845905 85...

4. The number e is irrational. Let us assume the opposite: let e be rational. a neutral number. Then there is such a large integer $N > 1$ what $N!e$ is an integer. But according to the previous one

$$\sum_{n=0}^N \frac{N!}{n!} < N!e < \sum_{n=0}^N \frac{N!}{n!} + \frac{2}{N+1}$$

All terms on the left are numbers $\frac{N!}{n!}$ integers, so there are two numbers

$$N!e \text{ and } \sum_{n=0}^N \frac{N!}{n!}$$

the difference between which is less than

$$\frac{2}{N+1} < 1$$

which is impossible.

Remark. The above proof of irrationality differences in the number e It is attributed to Fourier (1815) .1 Attempts have been made for a long time express the number e in terms of integers with the help of radicals or some other more algebraic operations. But it turned out that you can't specify an algebra- a mechanical equation with integer coefficients, the root of which would be the number of e's. The proof of this circumstance has been discovered comparatively recently.

1TO square the circle. With the attached issue

history compiled by F. Rudio. M.-L., 1936.P.

78.

Recently, in 1874, Charles Hermite (1822-1901); class-
 A mathematical proof can be found in Klein², which is updated in C.
 Langa³. A number that is not the root of an algebraic number.
 equations with integer coefficients are called transcendental. Number
 e was the first constant whose transcendence was strictly proved.
 zana.

The elements of the series for the number e are rational numbers, the series
 does not meet the Cauchy criterion, but its sum is not a rational number.
 scrap metal. This means that the Cauchy criterion is not valid for rational
 numbers. It follows that the proof of the Cauchy criterion for R is based on-
 The properties of the R field that are not present in the Q field are affected. At
 the end of the XIX century over the centuries, various theories of real numbers
 have been developed, in each of which in which some axiom is laid down, from
 which the Cauchy criterion follows. We'll take it as an axiom. Note only that
 from the algebraic point of view from this point of view, the statement about
 the existence of the number e is much more controversial. The first one is
 based on the proof of the existence of the number i. the completion time of the
 field Q, for which the Cauchy criterion is valid, which goes beyond algebra, but
 extends to a purely algebraic proof. proof of the existence of a quotient ring.

2.5. Power series as functions

Definition 4. Let be a power series

$$s = \sum_{n=0}^{\infty} a_n t^n$$

converges for certain numerical values of t forming a certain multiplicity of
 partnership G. We will say that this series defines a function whose domain
 is defined by which is divided by the set G: this function puts in the
 correspondence

² F. Klein Elementary mathematics from the point of view of higher education. Vol. 1.

³ Leng S. Algebra. Moscow: Mir Publ., 1967. Adding it.

point of sale $b \in G$ sum of the number series

$$\sum_{n=0}^{\infty} a_n b^n$$

which we will further denote as $s(b)$ or $s|_{t=b}$

We show that the series

$$\exp t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

we can consider as a function $\exp$ defined for all $t \in \mathbb{R}$ or even all of them $t \in \mathbb{C}$ By virtue of the Cauchy criterion, it is sufficient to prove that which is for any given value of t and any positive value ϵ There is an N such that the sums

$$\sum_{n=0}^M \frac{t^n}{n!} \quad (2.4)$$

by $M \geq N$ they differ from each other by no more than ϵ .

Now let t be an arbitrary real or even complex number. Then there is an integer K such that

$$K \geq |t| < K+1$$

let's put

$$q = \frac{|t|}{K+1} \quad 0 < q < 1$$

If $n > K$

$$\frac{|t|^n}{n!} = \frac{|t|^K}{K!} \frac{|t|^{n-K}}{n(n-1) \dots (K+1)} < \frac{|t|^K}{K!} q^{n-K}.$$

Let $M_2 > M_1 \geq N > K$ then the sums are

$$\sum_{n=0}^{M_1} \frac{t^n}{n!}$$

And

$$\sum_{n=0}^{M_2} \frac{t^n}{n!}$$

differ by no more than an amount

$$\sum_{n=M_1+1}^{M_2} \frac{t^n}{n!}$$

which we can estimate from above as

$$\frac{|t|^K}{K!} q^{M_1+1-K} + \dots + q^{M_2-K}$$

or, even more crudely, like

$$\frac{|t|^K q^{N-K+1}}{K! (1-q)}$$

Since as n grows, the expression q^n becomes smaller than any specified value numbers, you can always choose such a large number $N \geq K$ to

$$\frac{|t|^K q^{N-K+1}}{K! (1-q)}$$

it was less than the specified accuracy &. But then for any $M_1, M_2 > N$ times- the total number of the considered amounts does not exceed &.

Since in the series for hyperbolic and trigonometric functions, the same members, but possibly with different signs, are written out on the cards. the above estimates are also true for them. Therefore, the series $\exp t$, $\sinh t$, $\cosh t$, we can consider $\sin t$ and $\cos t$ as functions defined for all $t \in \mathbb{C}$

2.6. Arithmetic operations with power series

Let there be two power series, say,

$$f = \sum b_n t^n \quad g = \sum c_n t^n$$

and let them converge when $t = a$ As elements of a ring of formal powers, they can be added and multiplied. The values of $f(a)$ and $g(a)$ are the same you can add and multiply as elements of the field in question $\mathbb{R}$ or $\mathbb{C}$. It seems obvious that the substitution operation $t = a$ and arithmetic these actions can be reversed:

$$(f+g)|_{t=a} = f|_{t=a} + g|_{t=a}, (f \cdot g)|_{t=a} = f|_{t=a} \cdot g|_{t=a}$$

However, in reality, the second of these equalities can only be proved. under additional assumptions about the nature of convergence.

The problem is as follows. For anyone $\delta > 0$ of course, there is such an N that for every $M \geq N$ and $t = a$

$$|[f]_M - f| < \delta \quad [g]_M - g < \delta$$

where square brackets denote the truncation of the row for t^M It's not hard from here will receiveb assessment

$$|fg - f|_M |g|_V = |fg - fug + fug - fu|_M |g| \leq |g|^* + |f|^\delta$$

It is not difficult for a given task $\epsilon > 0$ select δ so that this value becomes less than a month. However, we are interested in

$$|fg - [fg]_M|$$

that is, we need to estimate the difference

$$[fg]_M - [f]_M [g]_M$$

which consists of members of the order $M+1, \dots, 2M$ Smallness of these terms It does not follow from the convergence of the f and g series.

Definition 5. It is said that a numerical series

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely if the series converges

$$\sum_{n=0}^{\infty} |a_n|$$

Of course, if a series with modules satisfies the Cauchy criterion, then it has a row without modules is also satisfied. Therefore, from the absolute convergence of the it blows normally, but not vice versa.

Theorem 10. Let be two power series

$$f = \sum b_n t^n \quad g = \sum c_n t^n$$

converge when $t = a$ absolutely. Then $f+g$ and $f \cdot g$ they also converge when $t = a$ absolutely.

$$(f+g)|_{t=a} = f|_{t=a} + g|_{t=a} \quad (f \cdot g)|_{t=a} = f|_{t=a} \cdot g|_{t=a}$$

The proof of this theorem is somewhat lengthy and is given, for example, in the lecture by A. Hurwitz⁴.

To our great joy in the proof of series convergence for $\exp t$, $\sinh t$, $\cosh t$, $\sin t$ and $\cos t$ each term of the series is evaluated modulo, so In fact, the absolute convergence of these series was proved.

2.7. Exponent Properties

The addition theorem was proved for the exponent as a power series

$$\exp(at) \exp(bt) = \exp((a+b)t)$$

The series $\exp(at)$ and $\exp(bt)$ converge absolutely at $t = 1$ By virtue of the theorem 10 artwork $\exp(at) \exp(bt)$ converges absolutely when $t = 1$ to the value $\exp((a+b)t)$ by $t = 1$

$$\exp(a) \exp(b) = \exp(a+b)$$

This is the addition theorem for the exponent in the usual school sense. All other properties of the exponent can be derived from it.

1. For everyone $a \in \mathbb{C}$ right

$$\exp(a) \exp(-a) = 1$$

2. For all real values of a , it is true

$$\exp(a) > 0$$

⁴ hUrwitz A. Teoriya funktsii [Theory of functions]. Ch. 1

By $a \geq 0$ all terms of the series $\exp a$ are positive and therefore their sum is the same. By $a < 0$ by virtue of n. 1

$$\exp(a) = \frac{1}{\exp(-a)} > 0$$

In particular, the equation $\exp(t) = 0$ it has no roots.

3. When $r \in \mathbb{Q}$ right

$$\exp(r) = e^r \quad (2.5)$$

For natural values of the variable t , this formula immediately follows n. 1:

$$\exp(2) = \exp(1+1) = e \cdot e = e^2, \dots$$

If $t = m/n$ where n, m are natural numbers, then a positive number $\exp(m/n)$ satisfies the equation

$$x^n = e^m$$

by virtue of n. 1:

$$\exp(m/n)^n = \exp(m/n + \dots + m/n) = \exp(m) = e^m$$

The positive root of this equation in elementary mathematics and both mean as $e^{m/n}$. Finally, if $t = -m/n$ then by virtue of n. 1

$$\exp(-m/n) = \frac{1}{\exp(m/n)} = \frac{1}{e^{m/n}} = e^{-m/n}$$

Thus, formula (2.5) is true for all rational signs of x .

Traditionally, formula (2.5) is considered to be valid for all those values of values of r , for which the series (2.1) can be used, including for all real objects. these values. In this case, the equality is

$$e^x = \exp(x)$$

taken as a definition.

Remark. In the future, there will be more than $e^{\sqrt{2}}$ but also e^A A , where A is a quad- military matrix. Of course, no one multiplies the number e A once.

2.8. Rounding error

If it is necessary to sum a large number of terms, the following occurs: each of the members turn into a decimal fraction and add up already decimals. This saves computer resources and time. However, the rounding error accumulates.

For example, we can calculate $\sin 32$ as the sum of 100 terms of a series for sinus:

```
sage: RR(sum([(-1)^n*32^(2*n+1)/factorial(2*n+1) for n in range(50)]))
```

0.551426671118916

87...

This answer is correct:

```
sage: RR(sin(32))
```

888

0.551426681241691

89...

However, if we convert the terms to decimals all at once, then ... -

chim

```
sage: sum([RR((-1)^n*32^(2*n+1)/factorial(2*n+1))
```

90...

```
for n in range(50)])
```

0.552008761698294

91...

The error in the 3rd decimal place is clearly visible. This error is not related to by truncating a row of one hundred members:

```
sage: sum([RR((-1)^n*32^(2*n+1)/factorial(2*n+1))
```

92...

```
for n in range(100)])
```

0.552008771821069

93 93

```
sage: sum([RR((-1)^n*32^(2*n+1)/factorial(2*n+1))
```

94...

```
for n in range(200)])
```

0.552008771821069

95...

This is a rounding error. Figure 2.1 shows that among the terms of the series there are

40...

very large numbers that also come with different characters.

Homework

Theoretical tasks

- 1) Define the convergence of a number series. What is the sum of num- row number?
- 2) Formulate the Cauchy criterion
- 3) Prove that $\sin 1$ is a real number.
- 4) Prove that $\sinh t$ converges for all real t .
- 5) Prove the basic trigonometric identity $\sin^2 a + \cos^2 a = 1$ for any real a
- 6) Prove that $|\sin a| \leq 1$ for any real a .
- 7) Prove the basic hyperbolic identity $\cosh^2 a - \sinh^2 a = 1$ for any real a
- 8) Prove that $e^i = \cos 1 + i \sin 1$
- 9) Prove that the number of terms in the series for $\exp t$ required for the calculation of using this expression when $|t| \leq 1$ with precision ϵ , does not exceed the number the terms required to calculate e with the same precision.

Practical tasks

- 1) Calculate the number e using a series with an accuracy of up to 10^{-9}
- 2) Calculate the number $\sin 1$ using a series with an accuracy of up to 10^{-5}
- 3) Calculate $e^{2.1}$ using the addition theorem $e^{2.1} = e^2 \cdot e^{0.1}$
- 4) Check the basic trigonometry by direct calculation of the series values. the classical identity for $t = 2$

- 5) Check by direct calculation of the values of the series the basic hyperbolic identity for $t = 2$
- 6) How many members do you need to take in the series for $\cos 32$ to get that the decimal fraction given by $\text{RR}(\cos(32))$?
- 7) (*) B Sage has a built-in $\sin$ function. Does the design allow you to $\text{RealField}(N)(\sin(1/10))$ calculate the value $\sin \frac{1}{10}$ accurate to rounding error, which is determined by the number of bits allocated to the is that a number?

Chapter 3.

Transcendental equations

3.1. Roots of the equation

Let f be a symbolic expression that depends on a single variable. Let x and at $a \leq x \leq b$ it takes real values. Allow-
stream that graph function $y = f(x)$ this segment can be used for the following tasks: with one movement of the hand, without lifting the pencil from the paper. Such a function it is called continuous.

From this point of view, it seems obvious that the following is true: the theorem.

Theorem 11. If the function f is continuous and on the segment $a \leq x \leq b$ a $f(a)$ and $f(b)$ have different signs, then on this segment the equation $f(x) = 0$ has at least one real root.

For example, the equation

$$x^2 = 2$$

has a single root on the segment $1 < x < 2$ as $f(x) = x^2 - 2$ accepts different signs for $x = 1$ and $x = 2$. This root, of course, is denoted as $\sqrt{2}$.

Quite a long time ago, a method was proposed that allows you to find approximate values of the roots of the equation whose existence is guaranteed by theorem 11 are given.

3.2. Dichotomy

The simplest way to find the root of an equation is

$$f(x) = 0$$

It is called the method of dividing a segment in half and consists of the following. -

putting it down

$$x_0 = a \quad y_0 = b$$

and divide this segment in half. If in the middle of the segment

$$x = \frac{x_0 + y_0}{2}$$

and on the left end $x = x_0$ if the function has different signs, then the equation has the root on the segment whose ends are

$$x_1 = x_0 \quad y_1 = \frac{x_0 + y_0}{2}$$

otherwise, the root is located on a segment with ends

$$x_1 = \frac{x_0 + y_0}{2} \quad y_1 = y_0$$

Dividing this segment in half again, we get the segment $x_2 < x < y_2$ length $\frac{b-a}{2^2}$ on which one of the roots The following estimate is obtained from the equation:

$$x_n \leq x \leq y_n \quad y_n - x_n = \frac{b-a}{2^n}$$

Thus, in n moves, we can find the root of the equation. before about $\frac{b-a}{2^n}$ By increasing n , we can calculate the root with any given point with a certain amount of information.

Listing 3.1. Calculating the root by dividing a segment in half

```
def dichotomy(f,a,b,epsilon):  
    if len (f. variables())==1:  
        x=f. variables ()[0]
```



```

else :
    return 'error'
if f. subs ([x==a])==0:
    return a
if f. subs ([x==b])==0:
    return b
if f. subs ([x==a])* f. subs ([x==b]) > 0:
    return 'error'
while abs(b—a)>2*epsilon:
    c=(a+b)/2
    if f. subs ([x==c])==0:
        return c
    if f. subs ([x==a])* f.subs ([x = c]) < 0
        b=c
else :
    a=c
return (a+b)/2

```

The implementation of the dichotomy method in Sage is presented in Listing 3.1. Arguments of the dichotomy function are f , the ends of the segment under consideration a and b and the accuracy ϵ with which to calculate the root of the equation $f(x) = 0$. We certainly don't keep the elements of the rows $x_0, x_1, \dots, y_0, y_1, \dots$, a note-
happy birthday x_n like a , a y_n like b . The calculation of the elements of these series goes up to those as long as the difference is $y_n - x_n$ it won't be less than 2ϵ . As soon as this difference is if the value is less than 2ϵ , the function returns a number $\frac{x_n + y_n}{2}$.

This number differs from the true root of the equation by no more than ϵ . In fact, by theorem 11, on the segment x_n, y_n the length of which is less than 2ϵ , there is a true root of the equation that differs from the middle this segment is no more than half its length, i.e. less than ϵ .

Example 5. For the equation $x^2 = 2$ on a segment $1 < x < 2$ accurate to $\epsilon = 0.001$ мы we have:

sage: load('sage/dichotomy.sage')	96 96
None	97...
sage: dichotomy(x~2-2,1,2,0.001)	98 98
1449/1024	99...
sage: RR(dichotomy(x~2-2,1,2,0.001))	100
1.41503906250000	101
sage: RR(sqrt (2))	102
1.41421356237310	103

It is clearly visible that we get the value $\sqrt{2}$ with an error in the 3rd character after with a comma.

The dichotomy method is very unpretentious and allows, in particular, to obtain approximate roots of algebraic equations of any degree. We can compare it with the standard one that we used many times in the course algebra classes.

Example 6. For the equation $x^5 + x + 1 = 0$ on a segment $-2 < x < 2$ with accurate to $\epsilon = 0.0001$ we have:

sage: f = x ~ 5+x+1	104
sage: RR(dichotomy(f, -2,2,0.0001))	105
-0.754821777343750	106
sage: RR(RR(dichotomy(f, -2,2,0.0001)))	107
-0.754821777343750	108
sage: QQ[x] (f).roots(AA,multiplicities=False)	109
[-0.7548776662466928?]	110

It is clearly visible that we get the value for the root with an error in the 5th character after the comma, the 4th one was determined correctly.

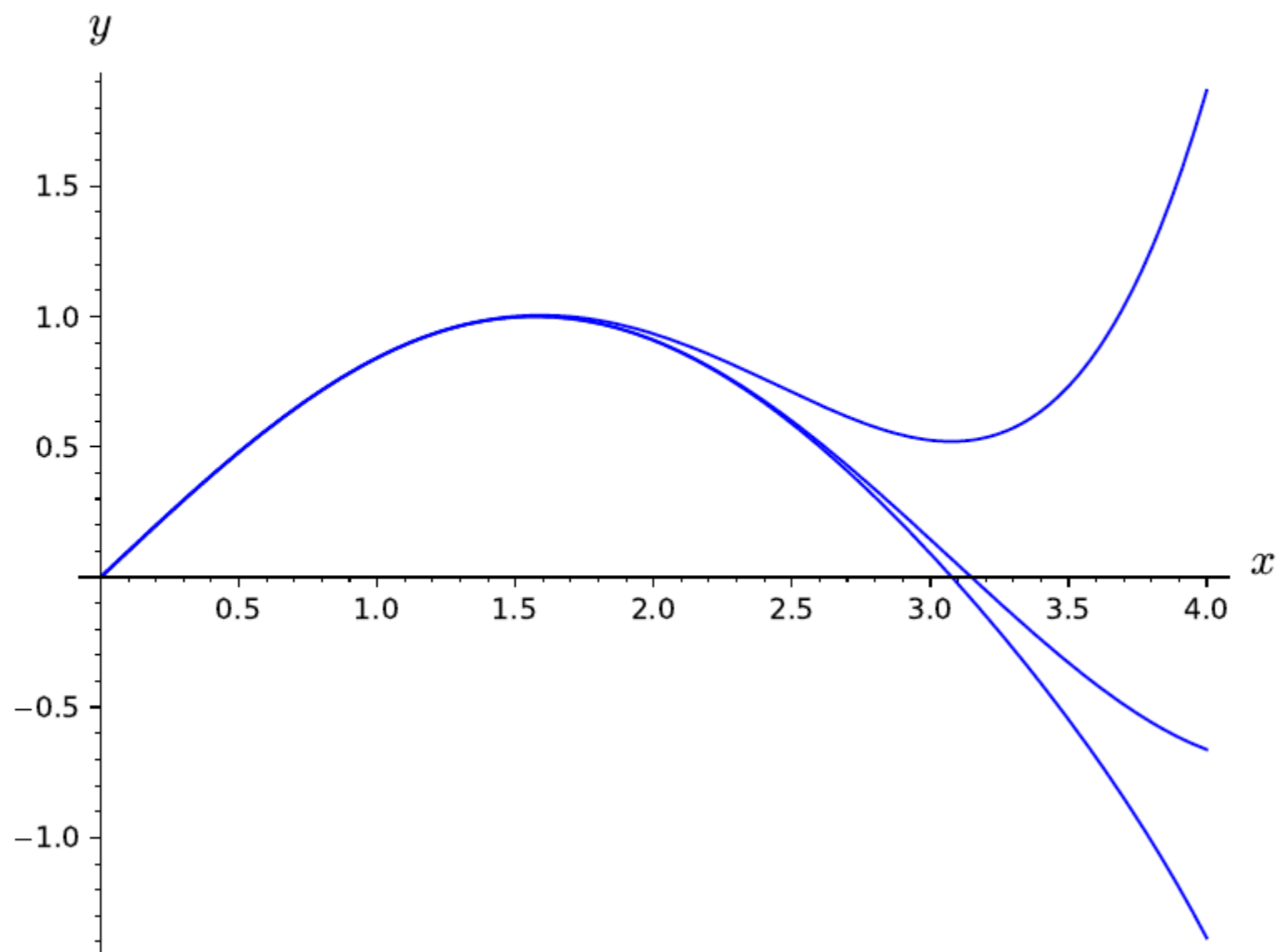


Figure 3.1. Graph of the sum of the first 3, 4, and 5 terms of the series for $\sin x$

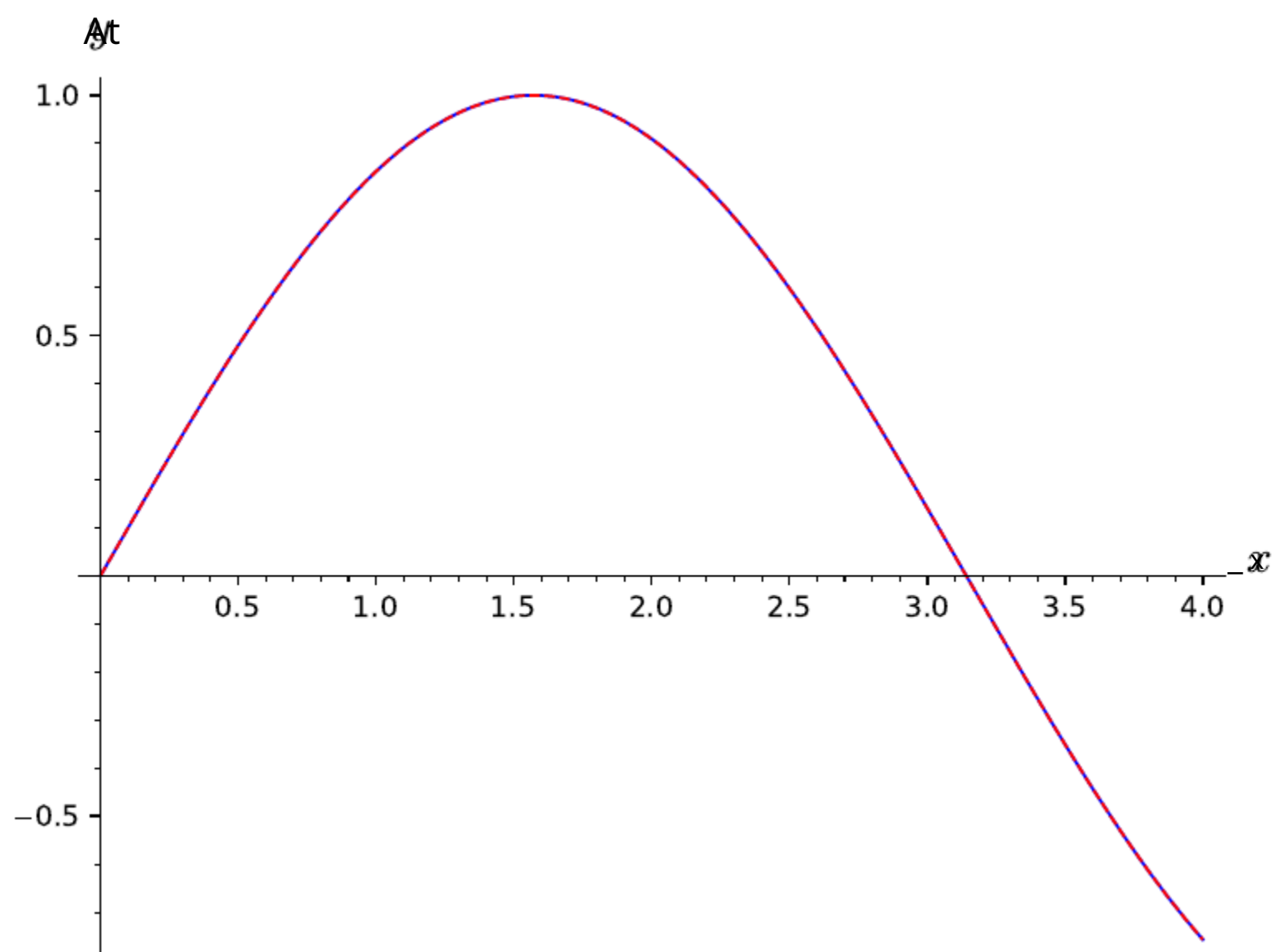


Figure 3.2. Graph $\sin x$ as the sum of 7 terms.

3.3. The number π

By definition, we assume that the number π is the smallest positive coefficient. ren equations

$$\sin x = 0$$

To calculate this number, plot the sine curve over the interval $0 < x < 4$

To this end, we recall that

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

taking 3, 4, and 5 terms in this series, we construct graphs of the resulting multi- For example, see Figure 3.1. The characteristic scale is already clear from this figure on the ordinate axis, all values lie between 1 and a small negative value. by number. Hence, if we calculate the sine values with precision $\epsilon = 0.01$ we won't be able to notice this error on the chart.

Let $M_2 > M_1 \geq N$ then the sum difference is

$$\sum_{n=0}^{M_1} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

And _

$$\sum_{n=0}^{M_2} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

one can very roughly estimate modulo as

$$\sum_{n=N+1}^{\infty} \frac{4^{2n+1}}{(2n+1)!}$$

and even rougher like

$$\sum_{n=2N+1}^{\infty} \frac{4^n}{n!} = e^4 \cdot \sum_{n=0}^{2N} \frac{4^n}{n!}$$

Knowing c is goodwith an accuracy of e , we can choose N such that this value was less than the one given to us $\epsilon = 0.01$

```
sage: RR(e~4-sum([4~n/factorial(n) for n in range
(13)]))
```

```
0.0149444321608598
```

```
sage: RR(e^4-sum([4^n/factorial(n) for n in range
(14) ] ) )
```

```
0.00416739027270552
```

This means that 7 terms will provide us with the required accuracy over the entire interval $0 < x < 4$

$$\sin x \approx \sum_{n=0}^7 (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

The graph of the polynomial on the right side is shown in Fig.3. 2 and it, with graphical accuracy, coincides with the graph obtained when using the sin function built into Sage. It is clearly seen that, first, such a schedule can be carried out without taking your hands off the paper, and, secondly, on a segment $3 < x < 4$ there is a root, which is called the number π .

We'll find it using the dichotomy method.

```
sage: f=sum([(-1)^n*x^(2*n+1)/factorial(2*n+1) for n in range (7)])
```

```
sage: RR(dichotomy(f,3,4,0.1))
```

```
3.187500000000000
```

```
sage: RR(dichotomy(f,3,4,0.01))
```

```
3.148437500000000
```

```
sage: RR(dichotomy(f,3,4,0.001))
```

```
3.141601562500000
```

```
sage: RR(dichotomy(f,3,4,0.000001))
```

```
3.14161396026611
```

It is not difficult to see that we have obtained with some accuracy the known value of π . However, an increase in the accuracy of the dichotomy does not lead to to refine the value. This is because the expression used is

for sine, it allows you to calculate the sine with the accuracy that we estimated in $\epsilon = 0.01$. If the values are $|f(x_n)|$ or $|f(y_n)|$ getting smaller $\epsilon = 0.01$ that sign $f(x_n)$ or $f(y_n)$ it may be defined with an error and we will choose another one. goy interval. Therefore, in this case, the calculations must either be stopped, or clarify the expression for the sine.

However, this observation casts doubt on the correctness of the application of applications of theorem 11 to the equation $\sin x = 0$. Where is the guarantee that when changing but a continuous smooth sine graph doesn't turn into something very intricately discontinuous? What you actually need to prove in order to use the dichotomy method without a twinge of conscience?

3.4. Justification of the method of dividing a segment in half

Let's start with the fact that in the prev. In this section, we have constructed two series of reals numbers

$$x_0, x_1, x_2, \dots, y_0, y_1, y_2, \dots$$

they are often called sequences of numbers. We hope that both will continue to sequences from the prev. the sections converge to the root of the equation in the sense of the next definition.

Definition 6. The set of points of a real line satisfying equal to the inequality $|x-a| < \epsilon$ It is called the ϵ -neighborhood of point a .

Denition 7. A point a of a real line is called the limit of investigative features $\{a_n\}$ if, outside any neighborhood of this point, there is only a finite number of points in the sequence is used. In this case, pi- jester

$$\lim_{n \rightarrow \infty} a_n = a \text{ or, in short, } a_n \rightarrow a$$

and imagine that with an increase of n points a_n . They're getting closer and closer to a or, in other words, how the sequence converges to a .

For example, that the sequence

$$a_n = \frac{(-1)^n}{n}$$

it has zero as its limit.

The definition does not say that the limit of a sequence can be only one, as it is easy to prove from the opposite. Let's assume that sequence $\{a_n\}$ has two different limits a and a' . These people have You can specify disjoint neighborhoods of U and U' for two points. As long as $a_n \rightarrow a$ only a finite number of elements lie outside of U . - and, since U and U' do not intersect, only these elements fall into U' . Therefore, the point a' has a neighborhood outside of which lies an infinite number of points. the number of elements in the sequence, which contradicts the definition the limit.

Arithmetic properties of the limit calculation operation can be collected into a single theorem.

Theorem 12. If

$$a_n \rightarrow a \text{ and } b_n \rightarrow b$$

TO _

$$a_n \circ b_n \rightarrow a \circ b$$

for any of the 4 arithmetic operations.

Proof. For example, we prove the statement of the theorem for multiplication of applications. No matter how small the number $\epsilon > 0$ we did not take all the elements of the sequence. government agencies a_n starting from a certain number, they fall into $|x-a| < \epsilon$ a all elements of the second sequence in $|x-b| < \epsilon$ so

$$|a_n b_n - ab| = |a_n(b_n - b) + (a_n - a)b| < (a+z)c + \epsilon(b+c) = (a+b+2c)c$$

For any neighborhood U of a point ab , we can choose a value so small that area

$$|x-ab| < (a+b+2\epsilon)\epsilon$$

it is placed in it entirely, and therefore, U contains all the elements after- satisfaction levels $a_n b_n$

□

Any numerical method for finding the root of an equation $f(x) = 0$ submit- it is a certain algorithm that finds the elements after- satisfaction levels $a_1, a_2, \dots$ and they hope that when replacing the root a with a_N no noticeable error is made if N is large. The easiest way the latter property is interpreted in the sense that

$$\lim_{n \rightarrow \infty} a_n = a$$

and $f(a) = 0$

That, that both sequences constructed by the dichotomy method are However, it is easiest to see from the trail. the principle.

Axiom 1 (on monotonic sequence). Monotonically increasing- the following upper-bounded sequence converges. Similarly, the mod- a notonically decreasing lower-bounded sequence converges.

Sequence $x_0, x_1, \dots$ monotonically increases:

$$x_{n+1} \geq x_n$$

and bounded from above:

$$x_n \leq b$$

So it has some limit with'. Similarly, the sequence $y_0, y_1, \dots$ monotonically decreases and is bounded from below by the constant a , so it has as its limit a certain number c

Limits with and c both sequences match. Then they have two disjoint neighborhoods. The first one contains all the points x_n start- the first one starts with a certain number, and the second one contains all the points y_n . But then the distance is between x_n and y_n all the time it remains more than a certain positive value. ranks are the distance between the right boundary of the first neighborhood and the left one border of the second one. This contradicts the fact that $y_n - x_n = \frac{|b-a|}{2^n} \rightarrow 0$

Thus, we prove that both sequences have the same and the same limit, denote it as C .

It remains to prove that this limit is the root of the equation $f(x) = 0$. From the con

$$f(x_0), f(x_1), \dots$$

they have a single sign, such as a plus sign. If the schedule is $y = f(x)$ conducted by one person by hand movement, from $x_n \rightarrow c$ should, what $f(x_n) \rightarrow f(x)$ So from $f(x_n) \geq 0$ follows $f(c) \geq 0$ For the sam

$$f(y_0), f(y_1), \dots$$

they have a different sign, so the limit is the sequence $f(c) \leq 0$ This means that $f(c) = 0$

Three important conclusions can be drawn from this.

First, the presented reflection provides a proof of the theorem 11. At the same time, we have again turned the ability to calculate something close to it is necessary to prove its existence.

Secondly, we proved the convergence of the dichotomy method under the conditions of the remy 11.

Third, we have seen how we can formulate the concept of continuous without referring to the process of plotting with your hands.

Definition 8. A function f is called continuous on the set M , if for any sequence $\{a_n\}$ from $a_n \rightarrow a \in M$ it follows that $f(a_n) \rightarrow f(a)$

Homework

Theoretical tasks

1) Formulate a definition of the sequence limit.

2) Formulate the definition of continuity of the function.

3) State the root theorem of the equation $f(x) = 0$

4) Describe the dichotomy method

Practical tasks

1) The equation is given

$$6x^5 - 75x^4 + 340x^3 - 675x^2 + 548x - 120 = 0$$

a) Find its roots in Sage using the root method.

b) Choose the interval $[a, b]$ so that you can find it using the dichotomy method the smallest root.

c) Choose the interval $[a, b]$ so that you can find it using the dichotomy method the largest root.

2) The equation

$$\prod_{n=0}^N (x-n) = 0$$

it has only whole roots. However,

```
sage: f=SR(ZZ[x](prod([x-n for n in range(30)]))) 124 f=SR(ZZ[x] sage: RR(dichotomy(f,1.0,10.0,10~-10
7.95803225371492126
```

Why is the error much larger than the specified accuracy? How is this implemented? vit?

3) Using the series for the exponent, find the root of the equation

$$e^x = 2$$

with accuracy $\epsilon = 0.01$ Compare with the exact solution $\ln 2$.

4) Using the series for the cosine, find the first positive root equations

$$\cos x = 0$$

with accuracy $\epsilon = 0.01$ Compare it with the exact solution.

54...

5) With what maximum accuracy can the number π be calculated as the number of roots of the equation $\sin x = 0$ using the approximation

$$\sin x \geq \sum_{n=0}^{10} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

We believe that (digits = 10^{-2}) gives the correct answer.

6) Is it possible to calculate π as the root of the equation $\sin x = 0$ with accuracy 10^{-10} if you use the built-in Sage sin function?

7) Modify the dichotomy function so that the loop stops when the accuracy of the calculation of f is exhausted.

8) We know that π is the smallest positive root of the equation $\sin \pi = 0$. We also know addition theorems for sine and cosine.
noosa. Prove that:

a) $\cos \frac{\pi}{2} = 0$

b) $\sin \pi n = 0 \quad n \in \mathbb{Z}$

c) $\sin (x+2\pi) = \sin x$

d) $\cos (x+2\pi) = \cos x$

9) (*) Prove that when $x \neq \pi n \quad n \in \mathbb{Z}$ the sine is not zero.